

1. Se $f: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ asociado a

$$M_B^B(f) = A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

$$B = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$M_C^C(f) = ?$$

$$M_C^C(f) = M_C^B(i_{\mathbb{R}^4}) M_B^B(f) M_B^C(i_{\mathbb{R}^4})$$

$$M_C^B(i_{\mathbb{R}^4}) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}_C = \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \delta \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{matrix} x = \alpha + \beta \\ y = \gamma \end{matrix}$$

$$y = \gamma$$

$$z = \gamma + \delta$$

$$t = -\alpha$$

$$\Leftrightarrow$$

$$\gamma = y$$

$$\alpha = -t$$

$$\beta = x + t$$

$$\delta = z - y$$

$$M_B^C(i_{\mathbb{R}^4}) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & +1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix}$$

$$M_C^C(f) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ -1 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 1 & -2 \\ 0 & -1 & 1 & 0 \\ 1 & -1 & 2 & 1 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

2. $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \rightarrow \begin{pmatrix} 2x_1 + x_3 \\ 2x_2 - x_3 \\ x_1 - x_2 + x_3 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda - 2 & 0 & -1 \\ 0 & \lambda - 2 & 1 \\ -1 & 1 & \lambda - 1 \end{vmatrix} = (\lambda - 2) [(\lambda - 2)(\lambda - 1) - 1] + (-1)(\lambda - 2) =$$

$$= (\lambda - 2) [\lambda^2 - 3\lambda + 2 - 1] = (\lambda - 2)(\lambda)(\lambda - 3)$$

$$\lambda_1 = 2 \quad m. e. 1$$

$$\lambda_2 = 0 \quad m. e. 1$$

$$\lambda_3 = 3 \quad m. e. 1$$

per il I criterio è
diagonalizzabile

$$V_{\lambda_1} = \left\{ \begin{array}{l} -z = 0 \\ z = 0 \\ -x + y + z = 0 \end{array} \right\} = \left\{ \begin{array}{l} z = 0 \\ x = y \end{array} \right\} = \left[\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right] \quad m. p. 1$$

$$V_{\lambda_2} = \left\{ \begin{array}{l} -2x - z = 0 \\ -2y + z = 0 \\ -x + y - z = 0 \end{array} \right\} = \left\{ \begin{array}{l} z = -2x \\ y = \frac{z}{2} = -x \end{array} \right\} = \left[\begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right] \quad m. p. 1$$

$$V_{\lambda_3} = \left\{ \begin{array}{l} x - z = 0 \\ y + z = 0 \\ -x + y + 2z = 0 \end{array} \right\} = \left\{ \begin{array}{l} x = z \\ y = -z \end{array} \right\} = \left[\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right] \quad m. p. 1$$

$$A = U \Lambda U^{-1} \quad \text{con} \quad \Lambda = \begin{pmatrix} 2 & & \\ & 0 & \\ & & 3 \end{pmatrix} \quad U = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 0 & -2 & 1 \end{pmatrix}$$

$$3. \quad B = \left\{ \underset{v_1}{(1, 1, 1)}, \underset{v_2}{(0, 0, 1)}, \underset{v_3}{(1, 0, 1)} \right\}$$

$$v_1' = v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \|v_1'\|^2 = 3$$

$$v_2' = v_2 - \frac{\langle v_2, v_1' \rangle}{\langle v_1', v_1' \rangle} v_1' = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$$

$$\|v_2'\|^2 = \frac{1}{9} + \frac{1}{9} + \frac{4}{9} = \frac{2}{3}$$

$$v_3' = v_3 - \frac{\langle v_3, v_1' \rangle}{\langle v_1', v_1' \rangle} v_1' - \frac{\langle v_3, v_2' \rangle}{\langle v_2', v_2' \rangle} v_2' =$$

$$= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{2}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{3 \cdot 2} \begin{pmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} 1 - \frac{2}{3} + \frac{1}{6} \\ -\frac{2}{3} + \frac{1}{6} \\ 1 - \frac{2}{3} - \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix}$$

base ortogonale $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix} \quad \|v_3'\|^2 = \frac{1}{2}$

base ortonormale $\begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{\sqrt{2}}{3} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$
 $v_1'' \quad v_2'' \quad v_3''$

$$v = (1, 5, -3)$$

$$v = \underbrace{\langle v, v_1'' \rangle}_{\alpha} v_1'' + \underbrace{\langle v, v_2'' \rangle}_{\beta} v_2'' + \underbrace{\langle v, v_3'' \rangle}_{\gamma} v_3''$$

$$\alpha = \frac{1}{\sqrt{3}} + \frac{5}{\sqrt{3}} - \frac{3}{\sqrt{3}} = \frac{3}{\sqrt{3}} ; \quad \beta = -\frac{1}{\sqrt{6}} - \frac{5}{\sqrt{6}} - 3\sqrt{\frac{4}{6}} = \frac{-6 - 3\sqrt{4}}{\sqrt{6}} = \frac{-12}{\sqrt{6}}$$

$$\gamma = \frac{1}{\sqrt{2}} - \frac{5}{\sqrt{2}} = -\frac{4}{\sqrt{2}}$$

$$4. \quad q(x, y, z) = x^2 + 2y^2 + z^2 + 2xz$$

$$|\lambda I - A| = \begin{vmatrix} \lambda-1 & 0 & -1 \\ 0 & \lambda-2 & 0 \\ -1 & 0 & \lambda-1 \end{vmatrix} = (\lambda-2)((\lambda-1)^2 - 1) =$$

$$= (\lambda-2)(\lambda^2 - 2\lambda + 1 - 1) = (\lambda-2)^2 \lambda$$

$$\lambda_1 = 2 \quad \text{m.a. } 2$$

$$\lambda_2 = 0 \quad \text{m.a. } 1$$

form quadratic
semidefinite positive

$$V_1 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \begin{array}{l} x - z = 0 \\ -x + z = 0 \end{array} \right\} = \left[\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right] \quad \text{m.g. } 2$$

$$V_2 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \begin{array}{l} -x - z = 0 \\ -2y = 0 \\ \underline{-x - z = 0} \end{array} \right\} = \left[\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right] \quad \text{m.g. } 1$$

base orthonormale $B = \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\}$

$$A = U \Lambda U^T = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 2 & & \\ & 2 & \\ & & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$